Gale Crater Diurnal Temperature Model: Plan Summary

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Goal and scope

The plan predicts ground and air temperature over a full diurnal cycle across all of Gale Crater, using Curiosity's REMS station as the only in-situ data source and physical terrain features as covariates, built in Python.

With a single station there is nothing to interpolate between, so the method is physics-anchored extrapolation: a physical baseline gives the diurnal shape at every pixel, and a learned model corrects it using REMS residuals and terrain covariates.

Item	Decision
Targets	Ground temperature and near-surface air temperature (modeled separately)
Time scope	Full diurnal cycle, sampled every 30 min of local true solar time, at selected seasons (Ls)
Spatial scope	Entire Gale Crater, starting at about 200 m grid spacing
Station data	REMS (Curiosity), already in hand
Validation	Held-out REMS sols, Curiosity traverse, and orbital thermal data
Tooling	Python

Data needed

Eight inputs cover the model: the station record, terrain, two surface properties, dust, the rover's path, and orbital thermal data for validation.

Data	Source	Used for
REMS air (ATS) and ground (GTS) temperature, LTST, Ls, quality flags	PDS Atmospheres volume mslrem_1001 (RMD .TAB/.LBL pairs)	Training targets, diurnal and seasonal calibration
Elevation, slope, aspect	Gale DTMs (CTX/HiRISE/HRSC, MOLA for context)	Terrain covariates, insolation
Thermal inertia	THEMIS / TES-derived maps	Main control on day-night temperature swing
Albedo	TES / MOC-derived maps	Absorbed solar energy
Dust opacity (tau)	Mastcam/REMS-era records, MCD dust scenarios	Time-varying, region-wide covariate
Curiosity traverse (position by sol)	MSL localization / traverse product	Spatial validation
Orbital surface temperature	THEMIS, TES (MCS for atmospheric profiles)	Spatial validation of ground temperature
Baseline climate	Mars Climate Database (`mcd` Python package)	Physical prior for the diurnal curve

Data processing

Processing turns raw station files and orbital rasters into one aligned, time-indexed dataset.

1. **Parse REMS.** Read each RMD .TAB with its .LBL label as the schema. Keep ATS, GTS, LTST, Ls, sol, and quality flags. Drop flagged or saturated records.
2. **Resample.** Bin to 5 to 15 minute intervals per sol and check which sols have continuous diurnal coverage.
3. **Build the grid.** Reproject all rasters to one coordinate system and grid (about 200 m to start) covering the crater and a margin.
4. **Derive terrain features.** Compute slope and aspect from the DTMs, then per-pixel insolation from slope, aspect, latitude, Ls, and LTST.
5. **Attach dust opacity** as a time-varying, region-wide value matched to each sol, not a raster.

6. **Encode time.** Use sine and cosine of LTST hour angle and of Ls. Never use UTC.
7. **Join station and covariates.** Extract covariate values at the rover's position for each sol, so the position changes as the traverse progresses.
8. **Match orbital observations** to model slices by LTST (within about 30 minutes) and Ls (within a few degrees).

Model stack

The model is a physics baseline plus a learned correction, so the diurnal shape comes from physics and the statistics only learn offsets.

```
T(x, t) = T_{base}(x, t) + f(\text{covariates}(x), \text{LTST}, L_s, \tau)
```

Layer	Method	Role
1. Physics baseline	Mars Climate Database, or a 1D thermal model (KRC or a custom heat-diffusion solver)	Diurnal curve at every pixel from thermal inertia, albedo, slope, elevation
2. Residual model	Gradient boosting (LightGBM or XGBoost) on baseline-minus-REMS residuals	Learn offsets from covariates, LTST, Ls, insolation, dust
3. Uncertainty (optional)	Gaussian process on residuals (GPyTorch or scikit-learn)	Prediction intervals and uncertainty maps
4. Residual kriging (fallback)	Kriging of leftover residuals	Only if validation shows spatial structure remains

Ground and air are separate models. Ground temperature is driven by thermal inertia, albedo, slope, aspect, and insolation, and varies strongly pixel to pixel. Air temperature is smoother in space, so it is modeled as a function of predicted ground temperature plus elevation and time of day. The ground-to-air gap is largest at night and in the afternoon, and it shifts with season.

Python stack: pandas and numpy for REMS, rasterio and xarray for rasters and the data cube, richdem or numpy gradients for terrain, `mcd` for the Climate Database, lightgbm, scikit-learn, and GPyTorch for models.

Validation

No single dataset validates everything, so the plan uses three layers, each with different strengths. The traverse matches what REMS measures but covers little ground. Orbital data covers the whole crater but only for ground temperature, at one local time per pass.

What is tested	Data	Variable
Time behavior (diurnal shape, seasons)	Held-out REMS sols and seasons	Ground and air
Spatial skill near the rover	Traverse: model prediction at the rover's position each sol vs. REMS	Ground and air
Spatial skill across the crater	THEMIS / TES at matching LTST and Ls	Ground only
Air temperature away from the rover	Not directly testable	Report as lower confidence

Rules for a fair test

- Split by whole sols or seasons, never by random rows, because neighboring readings are nearly identical.
- Match orbital observations on LTST and Ls, and check dust opacity at observation time.
- Avoid circularity: thermal inertia and albedo maps may be derived from the same orbital data, so validate against a held-out instrument or time window.
- Report orbital-vs-REMS measurement bias separately, since orbital brightness temperature is a pixel-scale skin estimate and REMS ground temperature is a small footprint.
- Report RMSE and bias by LTST bin, because night, dawn, and midday behave differently and one overall number hides problems.

Risks, open questions, next steps

The biggest risk is that one station constrains the model in one place only, so claims about the crater walls and Mount Sharp rest on the physics baseline and on orbital checks.

Risks

- Spatial skill is only directly testable along the traverse, and air temperature is not testable away from the rover.
- REMS air temperature is measured on the rover mast and can carry noise from the rover's own thermal environment.
- Orbital thermal data gives one local time per pass, so the diurnal cycle away from the rover is assembled from many passes.
- Dust storms shift temperatures strongly and can corrupt comparisons if opacity is not matched.

Open questions

- [ ] Which REMS sols are available (whole mission or a window)?
- [ ] Which thermal inertia and albedo products are available as rasters?
- [ ] Is the Curiosity traverse available as a file with coordinates by sol?
- [ ] Which seasons (Ls values) should the final map cover?

Next steps

1. Parse REMS into a clean per-sol table.
2. Run the physics baseline and compare it to REMS at the station.
3. Add the covariate stack and the residual model, then validate on held-out sols and the traverse.
4. Add the orbital check and, if time allows, the Gaussian process uncertainty layer.